

Basic EDA Concepts

Statistical Foundations, Data Preparation, and Exploratory Analysis Foundations

Part 1 of 2

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This reading (and this unit) has two parts.

Part 1 — today

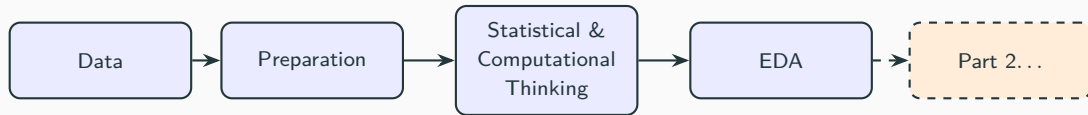
- Core statistical & probability concepts
- Data preparation and preprocessing
- EDA foundations: univariate and bivariate analysis

Part 2 — coming next

Multivariate/high-dimensional EDA, similarity & distance measures, feature selection, worked case studies, and responsible interpretation.

Why This Matters

KDD is often taught as a pipeline of stages — but every stage rests on the same small set of statistical and computational ideas.



- How missing values are handled changes what EDA can find.
- How a variable is measured decides which statistics are valid.
- Nearly every real dataset is a *sample* standing in for an unknown *population*.

Population, Sample, Parameter, Statistic

Population vs. Sample

- **Population:** the full group a question is about
- **Sample:** the subset actually observed

Parameter vs. Statistic

- **Parameter:** true population value (usually unknown)
- **Statistic:** sample-based estimate of it

Key idea

“95% of patients responded to treatment” is a statement about the *sample*. Whether it generalizes to the population depends on how representative that sample is.

Levels of Measurement

Four levels from measurement theory decide which operations on a variable are meaningful.

Level	Property	Example
Nominal	Categories, no order	Blood type
Ordinal	Ordered, no meaningful gap size	Low / Medium / High
Interval	Constant gap, no true zero	Temperature ($^{\circ}\text{C}$), calendar year
Ratio	True zero; ratios are meaningful	Weight, income, count

Why it matters

A 30°C day is 10 degrees warmer than 20°C — but *not* “50% warmer.” Celsius is interval, not ratio.

Descriptive vs. Inferential Statistics

- **Descriptive:** summarizes the data in hand (mean, median, variance, IQR)
- **Inferential:** uses a sample to draw conclusions about a population (confidence intervals, hypothesis tests)
- Variability is not noise to explain away — in most real data, it *is* the signal a KDD project is trying to characterize

Worked example

Six patients' annual medication costs: \$120, 140, 150, 160, 170, **3,200**.

- Mean $\approx$ \$656.70 Median = \$155

The mean suggests a “typical” patient pays over \$650/year — but five of six pay under \$200. One outlier distorts the mean, not the median.

Sampling Bias

Sampling bias: the sampling mechanism makes some parts of the population systematically more or less likely to appear.

- Hospital EHR data reflects patients who sought care there — not regional health generally
- A larger *biased* sample is still biased — more data does not fix it

Case study: telephone health surveys

The CDC's Behavioral Risk Factor Surveillance System is collected by phone. Households without a working phone, or who decline, are systematically excluded — so raw responses are reweighted before publishing population estimates.

Correlation and Association

- **Association:** any statistical relationship — knowing one variable tells you something about the other
- **Correlation:** specifically the strength/direction of a *linear* relationship (Pearson $r \in [-1, 1]$)

Common pitfall

A U-shaped relationship can yield $r \approx 0$ despite an obvious pattern. **Association without correlation** is common — and **correlation without causation** is even more common.

Probability Foundations

- **Random variable:** value is the outcome of a random process (e.g., # patients arriving per hour)
- **Expected value:** long-run average — the population-level analogue of the sample mean
- **Independence:** one event carries no information about another

Expected value, worked

Complaints per visit: 0, 1, 2, 3 with probabilities 0.4, 0.3, 0.2, 0.1.

$$E[X] = 0(0.4) + 1(0.3) + 2(0.2) + 3(0.1) = 1.0$$

Independence violated

Soil-moisture readings every 15 min are *not* independent — soil dries out gradually, so consecutive readings are highly correlated in time.

Bayes' Theorem & the Base-Rate Problem

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

Worked example: a screening test

Sensitivity 95%, specificity 90%, disease prevalence 2%. Patient tests positive — probability they're actually infected?

$$P(A | B) = \frac{(0.95)(0.02)}{(0.95)(0.02) + (0.10)(0.98)} \approx \mathbf{16\%}$$

Takeaway

A “95% accurate” test can still be *mostly wrong* when the condition is rare. Raw accuracy is meaningless without the base rate.

Computational Thinking in Data Analysis

Four pillars, each with a data-analysis analogue:

Decomposition Break a big question into smaller, answerable ones

Pattern recognition Reuse a technique from a similar past problem

Abstraction Decide which details matter for the question at hand

Algorithmic thinking Specify a repeatable, precise pipeline — not ad hoc edits

Key idea

Statistical thinking says what a finding *means*; computational thinking makes the analysis *reproducible and automatable*. A KDD project needs both.

Preprocessing I: Missing Values

Missing data is rarely safe to ignore. Three mechanisms:

Mechanism	Example	Implication
MCAR	Lab sample lost to shipping error	Deletion usually safe
MAR	Older patients less likely to have a wearable metric	Impute conditioned on age
MNAR	Sickest patients too unstable for a test	Deletion biases toward healthier cases

Common pitfall

Listwise deletion under MNAR doesn't shrink the dataset evenly — it systematically removes the sickest patients.

Preprocessing II: Encoding, Scaling, Leakage

Encoding & scaling

- One-hot: nominal categories
- Ordinal: order-preserving integers
- Min-max / standardization: comparable numeric scales

Integration & leakage

- **Tidy data:** variable = column, observation = row
- **Leakage:** using info unavailable at prediction time
- Leakage makes models look *better*, not worse — easy to miss

Case study

Diabetes 130-US Hospitals: patients appear across multiple encounters — splitting train/test by *encounter* instead of *patient* leaks information.

- **Univariate:** one variable's shape, center, spread, outliers
- **Bivariate:** relationship between two variables (scatter, grouped box plot, cross-tab)
- **Group-based:** comparing a statistic across categories — a direct line to the correlation-vs-causation caution ahead in Part 2

Anscombe's Quartet

Four datasets: nearly identical means, variances, and correlation — but visibly different relationships. **Plot before you trust a summary number.**

Summary & What's Next

1. Every dataset is a sample standing in for an unknown population.
2. Preprocessing decisions are analytical decisions.
3. EDA is a chain of questions — univariate and bivariate techniques are the building blocks.
4. Probability (especially the base-rate problem) is a practical, recurring tool.

Coming in Part 2

Multivariate & high-dimensional EDA ■ Similarity & distance measures ■ Feature selection ■ Full case studies ■ Responsible interpretation

Questions / Discussion